Ysbyty Gwynedd Anaesthetic Guidelines

ROTEM Familiarisation Guide

YG CG 502
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Key Points:

- **This is a Quick Guide for occasional users.**

Introduction

As part of the Obs Cymru project, a ROTEM Sigma machine (***Figure 1***) is available at Ysbyty Gwynedd.

Before reading on, watch this Introduction video: https://www.youtube.com/watch?v=Aq57mEY7PyQ

Why use it?

The use of ROTEM thromboelastography, facilitates targeted, individualised blood product administration. This would otherwise be empirical using standard lab based blood tests.

Figure 1: ROTEM Sigma



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When do I use it?

1. As part of the **Obs Cymru Project**: For all cases of measured blood loss> 1000ml
2. Non-Obstetric: Massive haemorrhage in theatres, critical care and emergency department.

Where is it?

The ROTEM Sigma machine is located in Maternity theatres.

Cartridges for Obstetric patients are located with the machine.

Cartridges for non-Obstetric patients are located in main theatres (next to ODP office).

How do I use it?

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- ROTEM must be used as part of a Major Haemorrhage protocol.
- Fill a clotting blood test bottle with 4.5ml of blood. Place a patient sticker on the bottle with the bar code visible.
- Ask an ODP (or midwife) if they are available to process the sample with ROTEM.
- Following processing, view the results of the ROTEM test remotely using the application, **Secure Viewer** (See attached pdf)**.**

Watch this short video on how to run a test yourself. https://www.youtube.com/watch?v=hmJDaXH6Rfk

How do I interpret the results?

*All you need is: **1** time value, **1** amplitude value and **2** ways to activate clot. That's it!*

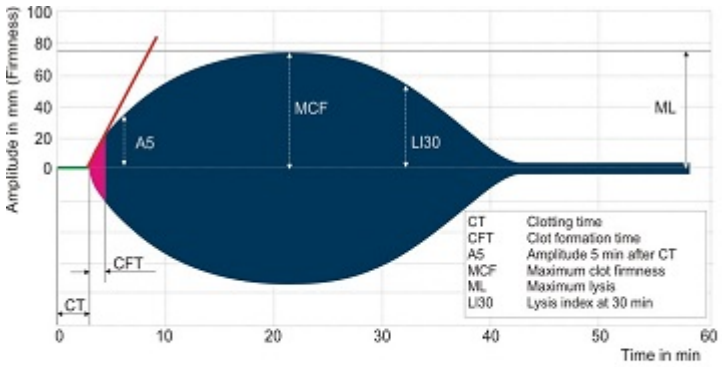


Figure 2:ROTEM Values

Values: (*Figure2*)

- **CT=Clotting Time**. It measures how long it takes for the clotting to be initiated (2mm amplitude). It is primarily affected by the levels of clotting factors present.
- **A5= Clot amplitude at 5 minutes after the clotting time**. It accurately predicts the final strength of the clot which is primarily determined by fibrinogen and platelet interaction.

Assays:

- **EXTEM**: by activating clot formation with thromboplastin (tissue factor), EXTEM gives an overview of the extrinsic pathway (factors VII, X, V, II, I), platelets and fibrinolysis.
- **FIBTEM**: differs from EXTEM by addition of a platelet-blocking agent. Consequently, FIBTEM looks at fibrinogen and fibrin polymerisation activity.

In order to determine what the patient requires, we only need to look at 3 values (***Figure 3):***

- **EXTEM CT**,
- **EXTEM A5** and the
- **FIBTEM A5.**

Detailed interpretation of results is not necessary. Just follow the algorithms within the major haemorrhage protocols (see below).

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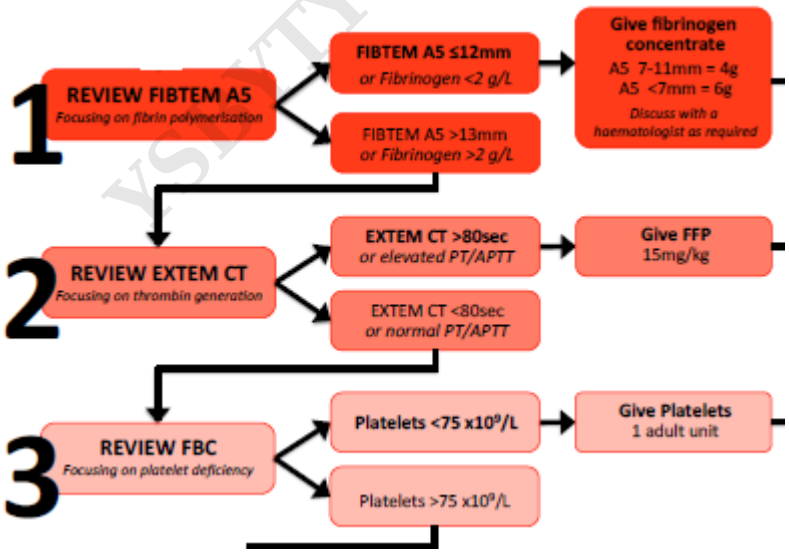
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Figure 3: ROTEM Result Page

Obstetric ROTEM Algorithm

N/B: This is for familiarisation only. Not to be used without referring to the full Obstetric Major Haemorrhage protocol.



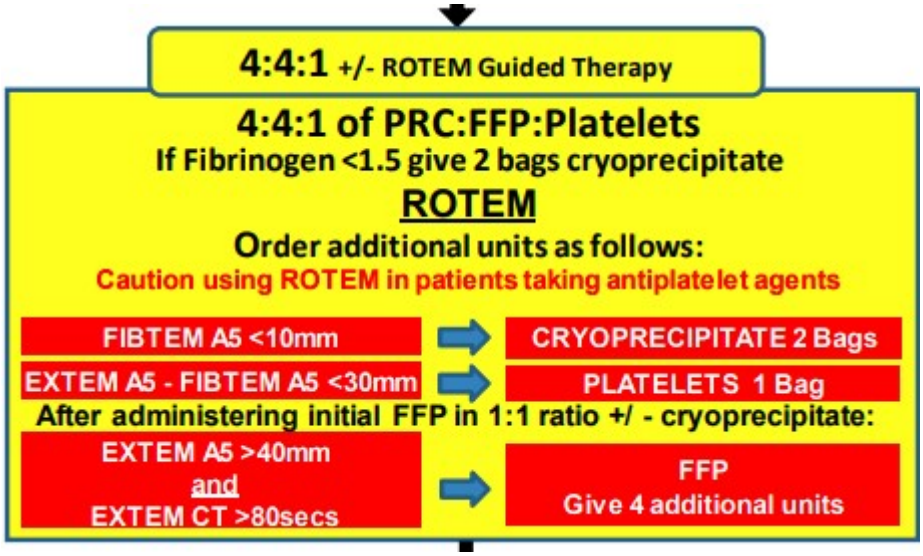
Non-Obstetric Algorithm

N/B: This is for familiarisation only. Not to be used without referring to the full Non-Obstetric Major Haemorrhage protocol.

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How do I find out more?

- 'ROTEM data interpretation teaching' pdf in the ROTEM folder in the FRCA drive.
- Obs Cymru Project: http://www.1000livesplus.wales.nhs.uk/obs-cymru

Ask Gareth Roberts or Stephan Clements

Keywords:
ROTEM, Haemorrhage
PDF:
ROTEM Familiarisation for New Starters.pdf

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